

Tong Chen

Head of Product Engineering & Founding AI Engineer @ Eigent | Agentic Systems · LLM Infra · Agent Evals | Singapore PR

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Agents will do the work. I build the worlds they work in.

EXPERIENCE

Eigent – Head of Product Engineering & Founding AI Engineer

Jan 2026 – Present · Singapore

Early core member of Eigent – open-source desktop multi-agent workforce built on CAMEL-AI (github.com/eigent-ai/eigent). I own the architecture end-to-end: took over the early codebase, led a ground-up refactor – agent runtime, backend, and deployment model – and now drive every layer of the system.

AGENT RUNTIME & STABILITY

- Long-horizon task reliability: checkpointed execution state, automatic recovery from mid-task failures (model timeouts, tool errors, crashes), idempotent retries – agents resume instead of restarting
- Real-world execution, not chat: agents control the browser, terminal, and file system end-to-end – with sandboxing, per-tool error containment, and human approval gates for destructive actions
- Concurrency: parallel multi-agent execution with inter-agent state coordination, backpressure on tool calls, context-window budget management
- Observability: every agent action, tool call, and artifact streamed live to the desktop client – full execution traces for debugging

ARCHITECTURE & INFRASTRUCTURE

- Dynamic task decomposition: planner builds a DAG of subtasks dispatched to specialized agents (browser, terminal, document, multimodal) in parallel
- One codebase, two deployments: fully local self-hosted (Docker, data on-device) and managed cloud – forcing clean abstractions at every layer
- Model-agnostic LLM gateway: routing across cloud APIs and local inference, with cost controls, fallback chains, abuse prevention
- Backend: Python/FastAPI, PostgreSQL, Redis – async task processing, credit/quota systems
- Mobile RL environments for agent training and evaluation

Design philosophy: the durable layer of the AI stack isn't the agent – it's the environment it runs in. Eigent's workspace is a configurable execution environment defining what agents can see, use, and act on.

ByteWizards – Founder & CTO

Dec 2023 – Present · Singapore

Founded ByteWizards to build AI-native products on one thesis: natural language becomes the interface, and agents do the work – software first, hardware next.

- xSale – domain agents for a vertical. Built a multi-agent system running cross-border influencer marketing end-to-end: AI-driven creator discovery and matching, automated outreach, and content generation on TikTok (approved TikTok Marketing API partner). Designed the agent pipeline and LLM workflow architecture; led a 7-person team from 0→1 to launch.
- NLUI – attacking the interaction layer. A mobile MCP host where users complete real-world tasks (ride-hailing, food delivery) through pure conversation, no app-switching. Designed the agent orchestration layer, tool-calling architecture, and mobile client. Established a research collaboration with Tsinghua University.

Shopee – Senior Full Stack Engineer

Apr 2019 – Nov 2023 · Singapore

Worked on high-performance tracking infrastructure and frontend platform engineering for large-scale e-commerce systems.

- Tracking Infrastructure: Architected Shopee's core tracking server and event pipeline services, supporting 1M+ QPS during the 12.12 peak traffic period with zero downtime. Managed event pipelines and services with a focus on scalability, reliability, performance optimization, and system stability.

- Frontend Team Leadership: Led a 5-person frontend team to build and maintain multiple data visualization platforms, including CampaignStation, BrandPortal, and Tidal, supporting business analytics, campaign operations, and internal decision-making workflows.
- Built internal UI libraries and monitoring systems to improve development efficiency, product consistency, platform maintainability, and operational visibility.
- Collaborated with backend, product, data, and business teams across infrastructure, frontend architecture, analytics platforms, and monitoring systems.

Zuzuche.com – Senior Frontend Engineer

Mar 2018 – Mar 2019 · Guangzhou

Led frontend development for web and mobile applications – user-facing features, reusable frontend infrastructure, and data tracking.

- Built reusable UI libraries standardizing frontend development across product pages
- Designed and implemented the data tracking system supporting user behavior analytics and business decision-making

Kingdee International Software Group – Senior Frontend Engineer

Jul 2016 – Mar 2018 · Shenzhen

Developed big data products and explored blockchain-related solutions at Kingdee Research Institute, focusing on enterprise software scenarios and data-heavy product experiences.

- Built frontend solutions for big data products, with a focus on data presentation, product usability, and enterprise application experience.
- Participated in blockchain solution exploration, technical research, prototype development, and early-stage product validation.
- Contributed to research-driven product development, working across frontend engineering, enterprise software, data visualization, and emerging technology exploration.

EDUCATION

Central China Normal University

Sep 2012 – Jun 2016

Bachelor of Science (Dual Degree), Mathematics & Applied Mathematics, and Physics

SKILLS

AI Agents

Multi-agent Systems

Large Language Models (LLM)

Software Architecture

Distributed Systems

Python

Go

Kafka

ClickHouse

Kubernetes

Docker

FastAPI

PostgreSQL

Redis

MCP

React